

Data Profile

This analysis draws on five linked tables released by JD.com for the 2020 MSOM Data-Driven Research Challenge, covering one consumable category over March 2018. Before any analysis, the raw data was profiled to establish its structure and to surface the anomalies that inform the cleaning steps that follow. The Raw data files are in the group "1_Raw"

Dataset Overview

The order table is the core, with 549,989 rows holding quantity, price, the four discount types, order type, promised delivery window, and origin and destination distribution centres. The delivery table (293,229 rows) carries the ship-out, station-arrival and final-arrival timestamps. The user table (457,298 rows) holds PLUS membership, city tier and demographics. The SKU table (31,868 rows) describes products, and the network table (56 rows) maps distribution centres to regions.

Field ▲	Mean	Median	Mode	StdDev	Min	Max
bundle_discount	0.63	0	0	6.44	0	252
coupon_discount	2.93	0	0	11.59	0	210
direct_discount	19.85	4	0	33.59	0	720
final_unit_price	71.57	58	0	75.88	-32	12158
original_unit_price	102.81	79	0	95.04	0	12158
quantity	1.22	1	1	1.84	1	400
quantity_discount	7.84	0	0	17.99	0	225

What the profiling showed:

Quantity is strongly right-skewed, with both a median and mode of 1, indicating that most records contain a single unit. Final unit price has a mode of 0 and a minimum of -32. Further inspection showed that all non-positive final-price records are flagged as gift items, indicating promotional or free-item accounting. These records were therefore retained where relevant to fulfilment but excluded from analyses of paid spending and promotions. The discount fields are mostly zero, with direct discounts appearing most prominently.

Cleaning Decisions:

The zero-price rows were checked against the gift-item flag, all 94,606 non-positive-price records (86,069 zero-price and 8,537 negative-price) are gift-item records, and were removed for spend-based analysis, leaving a cleaned base of 455,383 orders (the Orders_Cleaned query) as base data file Obj2 and Obj3 analyses.

Scope note:

Objective 1 is built from the raw order table (Obj1_fullfilment) rather than the Orders_Cleaned.csv base, because delivery reliability has nothing to do with price, and gift orders still carry deliveries and promised windows. Objectives 2 and 3, which analyse spend and quantity, descend from Orders_Cleaned file.

OBJECTIVE 1: FULFILMENT

Data preparation.

JD_order_data was inner-joined with JD_delivery_data on order_ID, producing 326,862 orders with delivery records.

Cleaning rationale

Delivery_hours ranged from -16.6 to 685.8 hours. Removed 5 records with delivery time <=0 and 1,793 records above 168 hours (7 days) as abnormal delivery records.

Final analysis base: 325,064 orders (1,798 rows; 0.55% removed).

Caveat: Some handling durations are slightly negative because dispatch times are recorded by the hour while order times include minutes. This affects 20,946 records (6.44%), all between approximately -0.93 and 0 hours, with none below -1 hour, consistent with timestamp granularity. These values were retained as rounding artifacts; across 325,064 records they do not materially affect group averages

1.1 How often is the promise met?

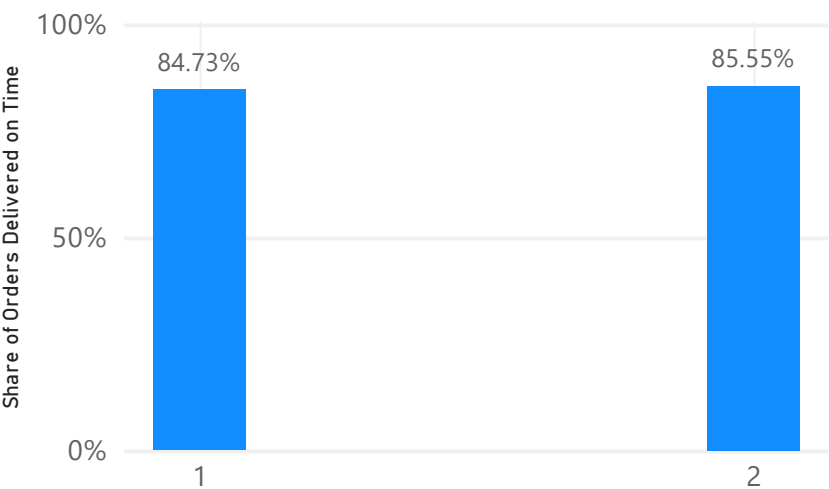
This checks whether JD's delivery reliability problem is linked to sales channel/type, especially whether 1P or 3P orders are weaker.

Findings:

JD meets the promise in 85% of orders. 1P and 3P are almost identical, so the issue is not channel-specific.

On-Time Delivery Rate: First-Party vs Third-Party Orders

Average of met by type. 1 = 'First Party, 2 = 'Third Party'



1.2 What does distance cost in time?

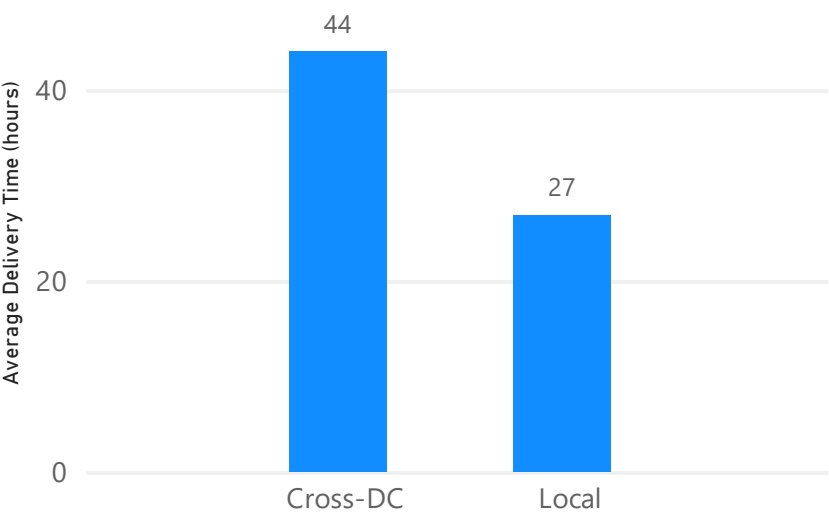
This tests whether orders shipped from a non-local DC take longer.

Findings:

Cross-DC orders take 44.0 hours on average, compared with 26.9 hours for local orders. The additional 17.1 hours means cross-DC delivery is about 64% slower. Since 38.8% of orders travel cross-DC, this route penalty affects a large share of volume.

Average Delivery Time: Local vs Long-Distance Orders

Timings of orders served from the local hub vs a distant hub



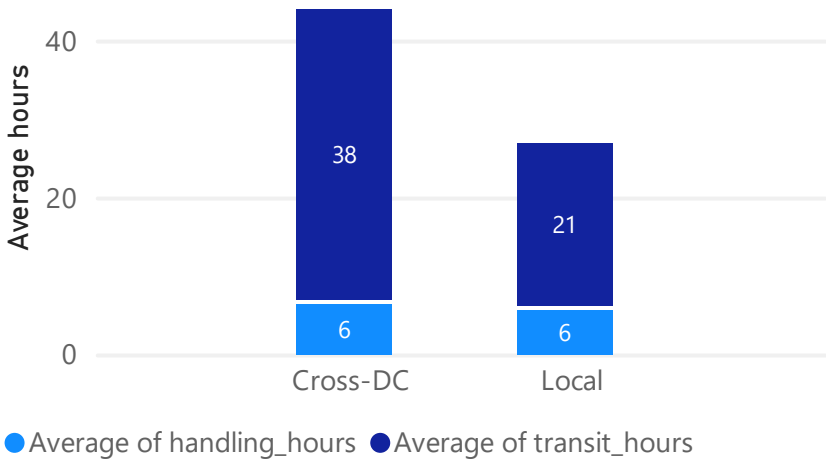
1.3 Is the delay warehouse or transport?

The solution depends on where the delay occurs.
Warehouse delay needs process improvement; transport delay needs network/routing improvement.

Findings:

Handling time is almost similar between local and cross-DC orders, around 5.7 to 6.5 hours. The real gap is in transit time, which rises from 21 hours to 38 hours. This shows that the problem is mainly distance and movement between locations, not warehouse handling.

Where Delivery Time Is Spent
Handling vs. transit times [Local vs cross DC]



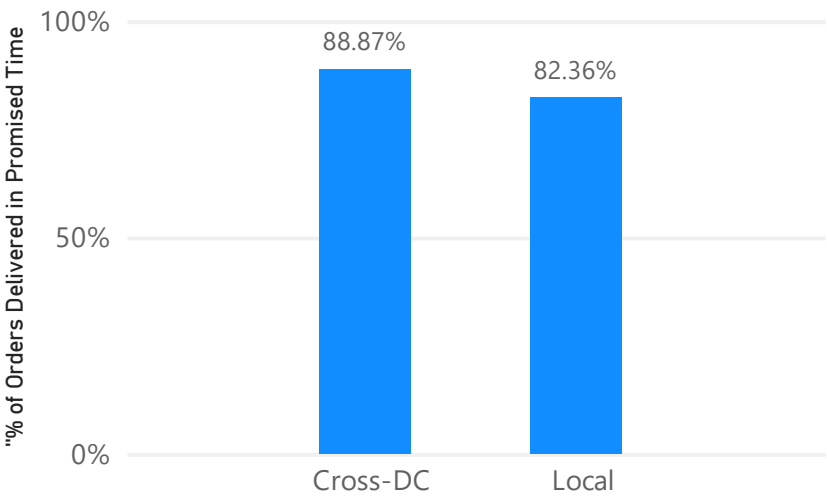
1.4 Do cross-DC orders miss the promise more often?

This checks whether longer deliveries directly translates into lower promise reliability.

Findings:

Surprisingly, cross-DC orders have better promise achievement at 88.9%, while local orders are lower at 82.4%. This is consistent with wider promised windows for Cross-DC records. The weaker area is local delivery reliability, where customers may expect faster delivery and the promised window is harder to meet.

On-Time Delivery Rate by Distance
On-time delivery share: Local vs. long-distance



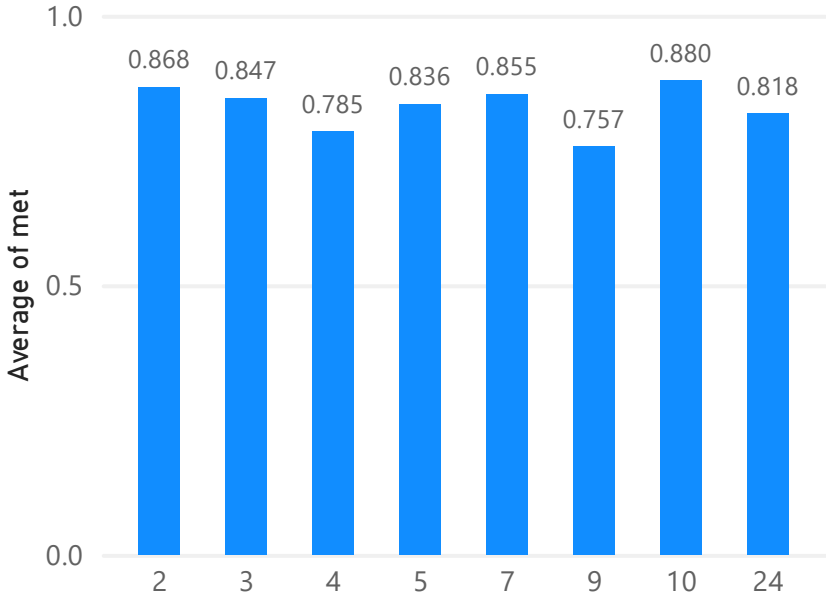
1.5 Which regions drag local reliability down?

Since Chart 1.4 shows local orders are weaker, this chart locates where the local problem is concentrated.

Findings:

Region 9 has the lowest local promise achievement at 75.7% across 34,883 orders, followed by Region 4 at 78.5% across 30,120 orders. Both regions have 8 DCs, so the weak performance does not appear to be simply because they lack DC capacity. Region 5 handles the highest load per DC, around 6,700 orders, but still performs better at 83.6%. This suggests Regions 9 and 4 are priority diagnostic locations.

Local Reliability by Region
Timings of orders served from the local hub vs a distant hub



OBJECTIVE 2: PROMOTIONS (1P)

Data preparation.

Built from Orders_Cleaned, the price-cleaned base of 455,383 orders. The data was filtered to first-party orders (type = 1), giving 259,049 orders. Third-party orders were excluded because their discounts may be designed or funded by independent sellers; this keeps the analysis focused on JD-controlled promotion mechanisms.

Derived fields.

Each order was classified by its discount combination: Direct, Quantity, Bundle, Coupon, None, or Multiple. Orders with more than one discount type were retained as Multiple rather than being forced into a single category. multi_unit flags orders with quantity above one unit. total_disc adds all four discounts, depth_pct calculates total discount as a percentage of original unit price, and depth_band groups discount depth into 0%, 0-10%, ..., and 50%+.

By grouping Discount type we get the following count: Multiple: 100819, Direct: 106786, Quantity: 23123, None: 20279, Bundle: 4328, Coupon: 3714

2.1 Which discount type is associated with higher order volume?

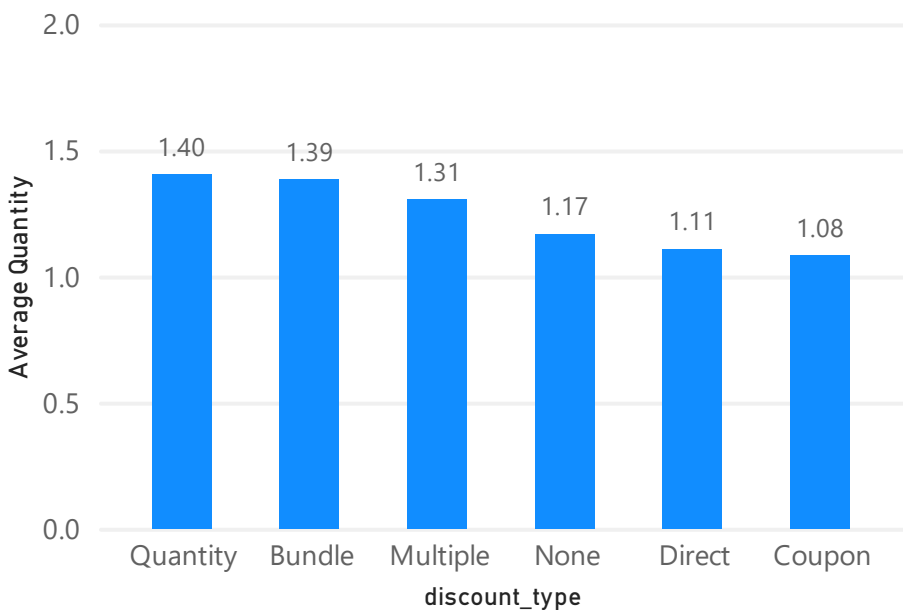
This chart compares average quantity across discount types against the no-discount baseline, showing which promotion mechanisms are associated with customers purchasing more units.

Findings:

Quantity discounts have the highest average quantity at 1.40 units per order, closely followed by Bundle discounts at 1.39 and Multiple discounts at 1.31. These all exceed the no-discount baseline of 1.17 units. In contrast, Direct discounts average 1.11 units and Coupon discounts average 1.08, both below the no-discount baseline. Direct discounts are the most common 1P discount type, yet they are not associated with higher order volume.

Average Units per Order by Discount Type

First-party orders only



2.2 Where does the higher volume come from?

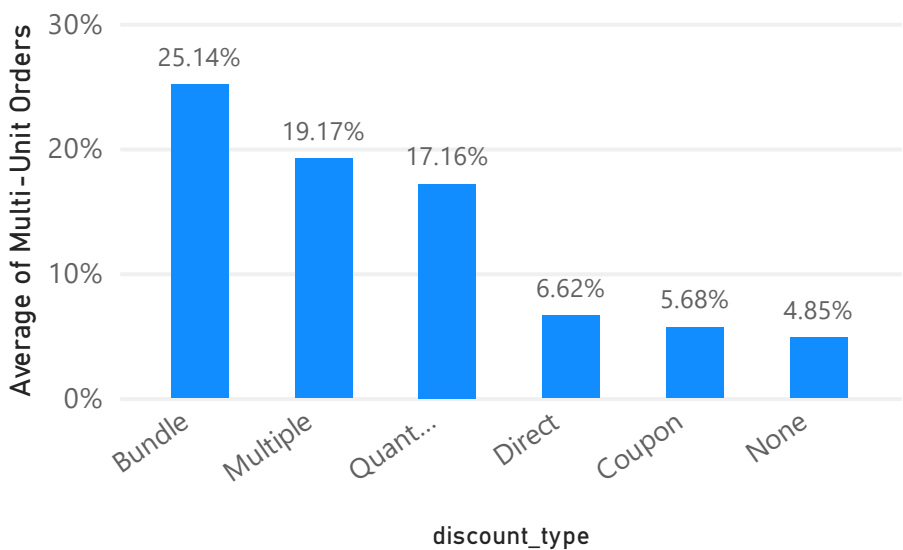
Since most orders contain one unit, this chart measures the share of orders with more than one unit to check whether higher average quantity reflects genuinely more multi-unit purchases.

Findings:

Bundle discounts produce are associated with the highest multi-unit order rate at 25%, followed by Multiple discounts at 19% and Quantity discounts at 17%. Direct discounts (6.6%) and Coupons (5.7%) are only slightly above the no-discount rate of 4.8%

Multi-Unit Order Rate by Discount Type

Share of orders containing more than one unit



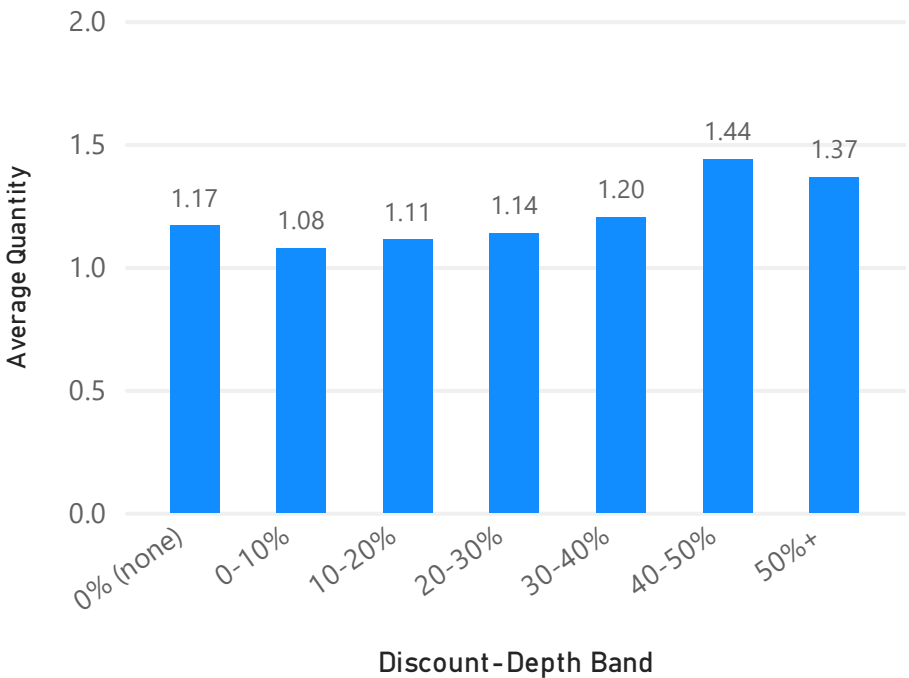
2.3 Does discount depth affect order volume?

This chart tests whether the size of the discount, regardless of its mechanism, is associated with higher quantity purchased per order.

Findings:

The pattern suggests a threshold rather than a steady increase. The 0% baseline averages 1.17 units, while shallow discounts from 0% to 30% remain at or below this level. Average quantity rises to 1.20 at 30–40%, then peaks at 1.44 for 40–50% discounts. It falls slightly to 1.37 for discounts above 50%. This suggests that shallow discounts may be insufficient to change purchase quantity, while discounts of around 40–50% are most strongly associated with higher volume.

Average Units per Order by Discount Depth



OBJECTIVE 3: MEMBERSHIP VALUE

Data preparation.

Built from Orders_Cleaned, the price-cleaned base of 455,383 orders. This table was merged with JD_user_data on user_ID to bring in the PLUS membership flag and city level. The join retained 455,383 orders.

Derived fields.

member_type was created from the plus flag: PLUS Member where plus = 1, and Ordinary otherwise. The existing spend field represents final transaction spend after discounts. city_level was retained for an optional geographic membership analysis.

Validation.

The final membership dataset contains 455,383 orders: 369,137 Ordinary orders and 86,246 PLUS Member orders.

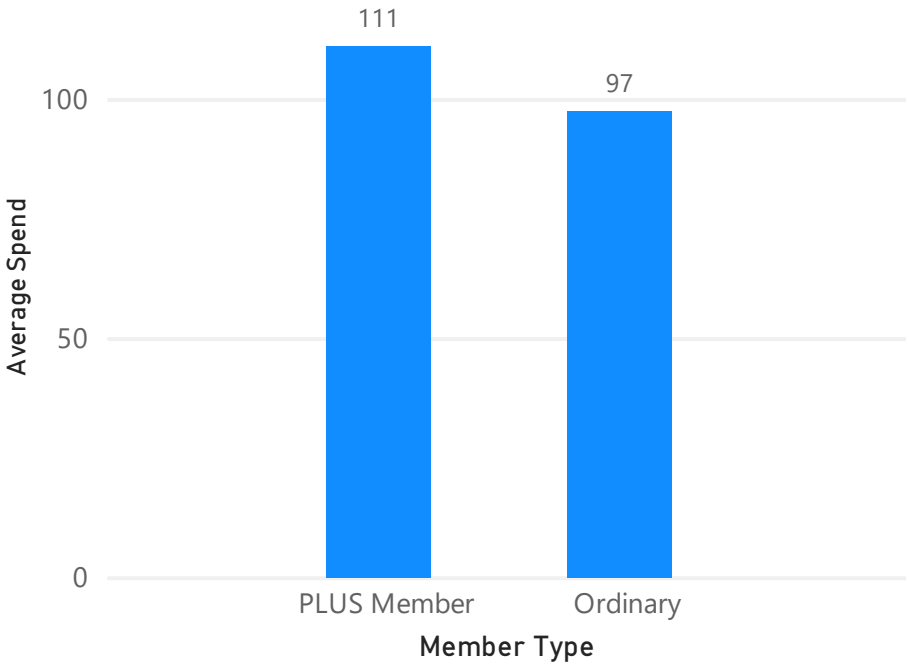
3.1 Do PLUS members spend more per order?

This chart compares average final spend per order between PLUS Members and Ordinary customers to establish whether membership is associated with higher customer value.

Findings:

PLUS Members spend an average of 111.11 per order, compared with 97.48 for Ordinary customers. This is a difference of 13.63, or approximately 14% higher spending per order. The result indicates that PLUS Members are associated with greater order value.

Average Spend per Order by Membership Type



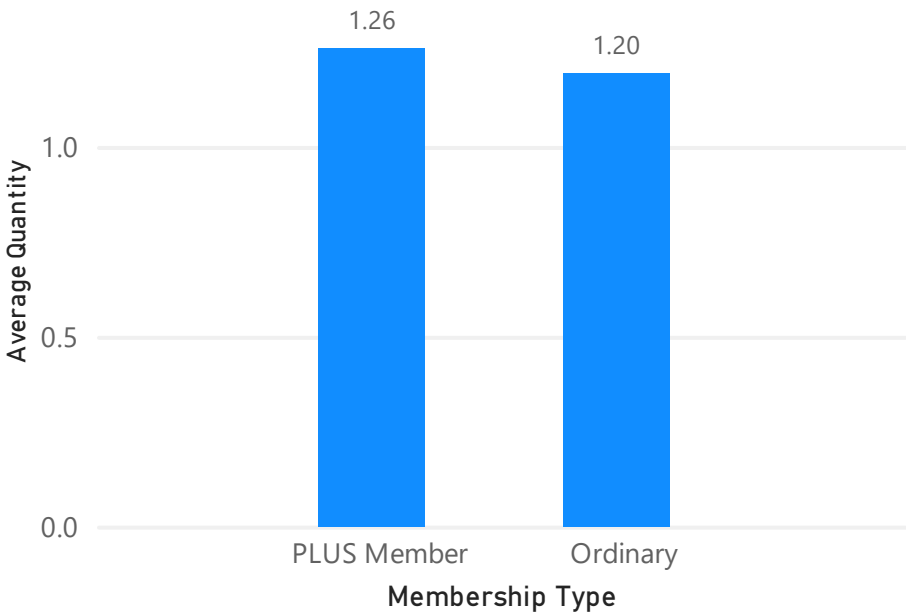
3.2 Do PLUS members buy more units?

This chart checks whether the higher spending of PLUS Members is also associated with purchasing more units per order.

Findings:

PLUS Members purchase an average of 1.261 units per order, compared with 1.196 for Ordinary customers. The difference is modest, around 5.4%, but is consistent with the higher spend result. Since the spend difference is larger than the quantity difference, PLUS Members may also be purchasing relatively higher-priced products; product mix would need further analysis to confirm this.

Average Units per Order by Membership Type
Comparison of PLUS and Ordinary customers



3.3 Where do PLUS members concentrate?

This chart compares the share of PLUS orders across city levels to identify whether membership participation differs by location type.

Findings:

PLUS membership share is highest in City Level 1 at 28.5%, then falls to 21.0% in City Level 2, 17.6% in City Level 3, and 15.0% in City Level 4. City Level 5 records 16.0%, but has a much smaller base of 5,911 orders. Overall, PLUS membership participation varies substantially by city level, with the strongest concentration in City Level 1.

PLUS Order Share by City Level

